Reparametrization

Def. Let $\alpha: I \to \mathbb{R}^3$ be a curve defined on an open interval I.
If $h: J \to I$ is a differential function on an open interval J,
then the composite $\beta = \alpha \circ h : J \to \mathbb{R}^3$ is a curve called a reparametrization of α by h.

Lemma. If β is the reparametrization of α by h, then the velocity is

$$\beta'(s) = \frac{dh}{ds}(s) \cdot \alpha'(h(s))$$

Proof. Denote $\alpha = (\alpha_1, \alpha_2, \alpha_3)$. Then, by the definition,

$$\beta(s) = (\alpha \circ h)(s) = (\alpha_1(h(s)), \alpha_2(h(s)), \alpha_3(h(s)))$$

Now, by the definition of velocity and the chain rule,

$$\begin{aligned}\beta'(s) &= (\alpha_1'(h(s)) \cdot h'(s), \alpha_2'(h(s)) \cdot h'(s), \alpha_3'(h(s)) \cdot h'(s)) \\ &= h(s) \cdot (\alpha_1'(h(s)), \alpha_2'(h(s)), \alpha_3'(h(s))). \\ &= h(s) \cdot \alpha'(h(s)).\end{aligned}$$

Thm. Given a regular curve α in $\mathbb{R}^3$,
there exists a reparametrization β of α such that $\|\beta'(s)\| = 1$ for all s.

Proof. Let $\alpha: I \to \mathbb{R}^3$ be a regular curve. Let $a \in I$ be fixed.
Define $S(t) = \int_a^t \|\alpha'(u)\|\, du$. Since $\|\alpha'\| \neq 0$, the $S(t)$ is strictly increasing on I.

Therefore, there is an inverse function $t: J \to I$ s.t $S(t(x)) = x$ for all $x \in J$,
where $J = S[I]$. Now, for $\beta = \alpha \circ t$,

$$\beta'(s) = \frac{dt}{ds}(s) \cdot \alpha'(t(s)) \quad \text{and} \quad \|\beta'(s)\| = \underbrace{\frac{dt}{ds}(s)}_{>0}\|\alpha'(t(s))\| = \frac{dt}{ds}(s) \cdot \frac{ds}{dt}(t(s)) = 1. \quad \blacksquare$$

Note: given a curve α, define length $S(t) = \int_a^t \|\alpha'(u)\|\, du$ and $\beta = \alpha \circ S^{-1}$.

「Def. Let $\beta: I \to \mathbb{R}$ be a unit-speed curve.
A vector field β' is called the unit tangent vector field on β, denote $T = \beta'$.
The derivative $T' = \beta''$ is called the curvature vector field of β.
The value $\kappa(s) = \|T'(s)\|$ $(s \in I)$ is called the curvature function of β.」

To restriction s.t $\kappa > 0$,
the unit-vector field $N = \frac{T'}{\kappa}$ is called the principal normal vector field of β.
The cross product $B = T \times N$ on β is called the binormal vector field of β.

「Lemma. Let β be a unit-speed curve in $\mathbb{R}^3$ with $\kappa > 0$,
Then, the three vector fields T, N, and B on β are unit vector fields that are mutually orthogonal at each point.
Proof. By definition, $\|T\| = \|\beta'\| = 1$. And since the curvature $\kappa > 0$, $\|N\| = \frac{1}{\kappa} \cdot \|T'\| = 1$.
And, since $\|T\|^2 = T \cdot T$ is constant, $2T \cdot T' = 0$, so $T \perp N$.
therefore the binormal, $\|B\| = \|T \times N\|^2 = \|T\|^2 \cdot \|N\|^2 - (T \cdot N)^2 = 1 - 0 = 1$. ∎

In summary, $T = \beta'$, $N = T'/\kappa$, $B = T \times N$, and $\{T, N, B\}$ is orthonormal.
Since $B \cdot T = 0$, $B' \cdot T + B \cdot T' = 0$ by differentiate. Therefore,

$$B' \cdot T = -B \cdot T' = -B \cdot \kappa N = 0$$

And since B is unit, $B' \cdot B = 0$, thus, there is a scalar value function τ s.t

$$B' = -\tau N$$

, called the torsion. Now

「Thm. Let β be a unit-speed curve in $\mathbb{R}^3$ with curvature $\kappa > 0$ and torsion τ,

$$\begin{aligned} T' &= \quad\ \kappa N \\ N' &= -\kappa T \qquad + \tau B \\ B' &= \quad -\tau N \end{aligned} = \begin{bmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{bmatrix} \begin{bmatrix} T \\ N \\ B \end{bmatrix}$$

Proof. The first and third rows are just definition. To show the second, observe
$N' = (N' \cdot T)T + (N' \cdot N)N + (N' \cdot B)B$
Now, $N' \cdot T = N \cdot T'$ and $N' \cdot B = N \cdot B'$ gives the results.

「Prop. Let β be a unit-speed curve in $\mathbb{R}^3$ with $\kappa>0$. Then,

β is a plane curve if and only if $\tau=0$, the torsion is zero.

Proof. Suppose that β is a plane curve. That, $\mathrm{Im}\,\beta$ lies in some plane.
Therefore, there are two vectors $p, q \in \mathbb{R}^3$ such that $q \neq 0$ and

$$[\beta(s)-p]\cdot q = 0 \text{ for all } s.$$

Differentiating yields:

$$\beta'(s)\cdot q = 0 \quad \text{and} \quad \beta''(s)\cdot q = 0 \text{ for all } s.$$

That is, q is orthogonal to $T=\beta'$ and $N=\beta''/\kappa$. So, B and q are parallel,
we can conclude that $B=\pm q/\|q\|$. So $B'=0$, the torsion τ must be zero.」
Conversely, suppose that the torsion is zero. Then, $B'=0$ by definition.
($B'=0$ means B is parallel.) To show that $\mathrm{Im}\,\beta$ lies in some plane, define

$$f(s)=[\beta(s)-\beta(0)]\cdot B(s) \text{ for all } s.$$

Then, the derivative is

$$f'(s)=\beta'(s)\cdot B(s)+\beta(s)\cdot B'(s)=\beta'(s)\cdot B(s)\overset{\text{def.}}{=}T\cdot B=0.$$

By definition, $f(0)=0$, so $f=0$. Consequently,

$$[\beta(s)-\beta(0)]\cdot B = 0 \text{ for all } s,$$

So proved ☒

「Prop. Let β be a unit speed curve with constant curvature $\kappa>0$ and torsion zero.
Then, the β is part of a circle of radius $1/\kappa$.
Proof. Since $\tau=0$, the β is a plane curve. Define a curve

$$\gamma(s)=\beta(s)+\frac{1}{\kappa}\cdot N(s).$$

By the Frenet formular, $\gamma'=\beta'+\frac{1}{\kappa}\cdot N'=T+\frac{1}{\kappa}\cdot(-\kappa T)=0$. Therefore, γ is constant.
Now, $d(\gamma,\beta(s))=\|\gamma-\beta(s)\|=\frac{1}{\kappa}\cdot\|N\|=\frac{1}{\kappa}$, so proved.

「General Version of Frenet formula」

Let $\alpha: I \to \mathbb{R}^3$ be any regular curve, not necessary $|\alpha'| \equiv 1$.
And, let $\beta: J \to \mathbb{R}^3$ be a reprametrization of $\alpha[I]$ by the arc-length

$$s = s(t) = \int_{t_0}^{t} |\alpha'(u)|\, du, \quad \text{where } t_0 \in I.$$

with the inverse function $t = t(s) = s^{-1}$. Then,

1). $\dfrac{dt}{ds} = \dfrac{1}{|\alpha'|}$ and $\dfrac{d^2t}{ds^2} = -\left(\dfrac{\alpha' \cdot \alpha''}{|\alpha'|^4}\right)$

2). The curvature and torsion of α is given by

$$\kappa(t) = \frac{|\alpha' \times \alpha''|}{|\alpha'|^3} \quad \text{and} \quad \tau(t) = -\frac{(\alpha' \times \alpha'') \cdot \alpha''}{|\alpha' \times \alpha''|^2}.$$

Proof. Since $\dfrac{dt}{ds} = \left(\dfrac{ds}{dt}\right)^{-1} = \left(\dfrac{d}{dt}\int_{t_0}^{t} |\alpha'(u)|\, du\right)^{-1} = \dfrac{1}{|\alpha'|}$, the first is obtained

And, $\dfrac{d}{ds}\dfrac{dt}{ds} = \dfrac{d}{ds}\left(\dfrac{1}{|\alpha'(t)|}\right)$

「Signed Curvature for plane curve.

Let $\alpha : I \to \mathbb{R}^2$ be a plane curve.

Then, the binormal B is constant vector, so we consider only T and N.